

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD organic/materials chemist with 6+ years in synthesis, purification, analytical characterization, and DOE-guided process development. Builds reproducible process windows and technical data packages connecting reaction/process variables to material quality, stability, robustness, and scale-up decisions.

CORE COMPETENCIES

Small-Molecule Process Development

DOE / Process Characterization

Organic Synthesis

Crystallization Mindset

Scale-Up Documentation

cGMP/GDP Readiness

Analytical Characterization

QbD Mindset

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided chemical process-development experiments for bio-based polymer systems, defining process windows, analytical readouts, reproducibility criteria, and SOP-ready scale-up specifications.
- Translate process variables, FT-IR/thermal/spectroscopic data, and material-quality outcomes into technical recommendations and repeatable procedures.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, purified, and characterized 15+ organic electrochromic material variants, connecting molecular structure, purity, stability, and performance with NMR, GPC, MS, FT-IR, UV-Vis-NIR, TGA/DSC, and electrochemistry.
- Diagnosed degradation and failure modes through hypothesis-driven experiments, analytical comparisons, and structured datasets supporting reaction/process iteration.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded process development for molecular thin-film coatings, optimizing surface preparation and CVD conditions to improve reproducibility and optical/material performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition under flow using DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and process controls.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30% and strengthened data integrity.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

DOE-Guided Process Development Current

Built process windows, analytical readouts, reproducibility criteria, and SOP-ready specifications for bio-based polymer scale-up.

DOE, SPC, FT-IR, thermal analysis, process variables

Organic Synthesis and Analytical Characterization Chemistry of Materials 2025

Synthesized and purified molecular variants, then connected structure, purity, stability, and performance through multi-technique data.

Organic synthesis, NMR, MS, GPC, FT-IR, UV-Vis-NIR, CV

Process-Controlled Thin-Film Coatings DoD Funded

Developed CVD/surface-preparation strategies and translated process conditions into reproducible material outputs.

CVD, QCM-D, surface preparation, process optimization

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process & Reaction Development: DOE, process characterization, route/reaction analysis, robustness/productivity studies, scale-up specifications, SOPs, QbD/cGMP/GDP readiness

Analytical Characterization: NMR, LC-MS exposure, GC-MS, GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry, chromatography-adjacent workflows

Chemistry & Materials: Organic synthesis, purification, crystallization concepts, polymers, small-molecule materials, structure-property-process analysis

Data & Documentation: SPC, Excel, Python/basic, JMP/basic, electronic/structured lab records, technical reports, publication-quality writing

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.

3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)